ALY 3510

Midterm Study Guide

AI Foundations: An Interdisciplinary Approach

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UPDATED WITH THREE RECOVERED QUIZZES + MIDTERM FORMAT

Midterm Exam Snapshot

Exam	Online midterm, multiple-choice / quiz-like
Questions	30 total
Time	60 minutes
Points	100 total; questions have variable point values
Availability	October 26, 12:00 a.m. through October 30, 11:59 p.m. EST
Delivery	Respondus LockDown Browser + Respondus Monitor
Attempts	The course guidance says unlimited attempts, while also saying the exam can be taken only once. Treat the one-exam restriction as the safe assumption unless Canvas shows otherwise.
Practice	A 5-point Respondus Monitor Practice Quiz is available before the exam

Practical implication: This is a recognition-and-recall exam under time pressure: about 2 minutes per question on average. Practice should emphasize quick discrimination between similar concepts and "Select ALL that apply" wording rather than long-form writing.

Respondus Checklist

- Use a quiet, distraction-free location where you are comfortable being recorded.
- Turn off and move other devices out of reach.
- Clear books, papers, and other unpermitted materials from the workspace.
- Stay at the computer for the duration of the test.
- If the computer, Wi-Fi, or location changed since your prior check, rerun the System & Network Check and Webcam Check.
- Use a firm desk/table, good front lighting, and avoid hats with brims or backlighting.
- Complete the 5-point Respondus Monitor Practice Quiz before the real exam.

How to Practice This Week

- Use 30-question mock exams to mirror the real exam length.
- Use a 60-minute timer. For drills, target 90–120 seconds per question.
- Mix single-answer and Select ALL that apply questions.
- After each mock, sort misses by concept, not just by question number.
- Re-test missed concepts in a later mock with different wording.
- Do at least one closed-book mock under Respondus-like conditions: clear desk, no phone, no notes, one sitting.

Core Review

AI foundations: Alan Turing and the Turing Test; Dartmouth Conference (1956); symbolic AI, machine learning, and neural networks; AI Winter; strengths of human versus artificial intelligence.

Knowledge representation: Semantic networks represent relationships as graphs. Frames organize knowledge into structured packages, often with slots and default values.

Reasoning: Deductive: rule to necessary conclusion. Inductive: examples to probable generalization. Abductive: observation to best explanation. Analogical: reason from similarity. Predictive: estimate future outcomes.

Machine learning: Supervised learning uses labeled examples; unsupervised learning finds structure without labels; reinforcement learning learns through rewards/penalties. Know classification vs. regression, training/validation/test data, overfitting, underfitting, and generalization.

Neural networks / deep learning: Inputs are combined through learned weights and nonlinear activations. Training uses a loss signal and weight updates. Deep learning learns layered representations rather than relying on explicitly programmed rules.

Human vs. machine learning: Humans are generally more sample-efficient: people can often learn meaningful concepts from few examples while many ML systems require much more data.

Ethics and society: Accountability, opacity/explainability, and bias/fairness matter when neural systems affect people. Adaptive machines also raise questions about human agency, education, work, and expertise.

Emergence and cognition: Complex behavior can emerge from interactions among simpler components. Deep learning's success raises interdisciplinary questions about pattern recognition, mathematical optimization, cognition, and intelligence.

Distributed AI, frameworks, and NLP: Distributed systems spread intelligence or computation across agents, devices, or organizations. Study emergence without centralized control, shared accountability, data ownership/privacy, shifting power, AI-framework accessibility, and NLP questions about statistical pattern recognition, language inequality, authorship, misinformation, and whether effective communication requires meaning or intention.

Recovered Quiz 1: Due September 27

Status: Seven verbatim questions were recovered from the September 26 conversation. The quiz likely contained additional questions, but they were not recoverable from the history excerpts, so they are not presented as originals here.

1. Who is widely considered to be the "father of artificial intelligence" and proposed the famous test for machine intelligence?

- John McCarthy
- Marvin Minsky
- John von Neumann
- Alan Turing

Answer: Alan Turing

2. Which of the following are major AI paradigms or approaches? (Select ALL that apply)

- Neural Networks
- Machine Learning
- Quantum Computing
- Symbolic AI (Knowledge-based systems)

Answer: Neural Networks; Machine Learning; Symbolic AI (Knowledge-based systems)

3. Which conference is considered the official founding event of AI as a field of study?

- The Macy Conferences (1940s–1950s)
- The First International Joint Conference on AI (1969)
- The Dartmouth Conference (1956)
- The MIT AI Laboratory founding (1959)

Answer: The Dartmouth Conference (1956)

4. Which factors contributed to the "AI Winter" periods? (Select ALL that apply)

- Overpromising and underdelivering on AI capabilities
- Lack of interest from computer scientists
- Reduced government and private funding
- Limited computational power and memory

Answer: Overpromising and underdelivering; reduced government and private funding; limited computational power and memory

5. In knowledge representation, what is a "semantic network"?

- A neural network trained on semantic data
- A computer network that processes natural language
- A database of word meanings and definitions
- A graph structure representing relationships between concepts

Answer: A graph structure representing relationships between concepts

6. Which of the following is a strength of human intelligence compared to AI?

- Speed of calculation
- Emotional understanding and empathy
- Ability to process vast amounts of data quickly
- Consistency in decision-making

Answer: Emotional understanding and empathy

7. What is the main advantage of using frames in knowledge representation?

- They eliminate the need for logical reasoning
- They enable faster mathematical computations
- They automatically generate new knowledge
- They organize knowledge into structured packages with default values

Answer: They organize knowledge into structured packages with default values

Recovered Quiz 2: Due October 11

Status: Seven verbatim questions and one additional recovered concept/answer were found in the October 9 conversation. The prompt for the sample-efficiency item was truncated, so it is kept separate from the verbatim questions.

1. What ethical concerns arise when neural networks make decisions that affect human lives? (Select ALL that apply)

- Accountability becomes unclear when decisions are made by algorithms rather than humans
- Neural networks process information too quickly for human oversight
- The decision-making process may be opaque and difficult to explain to those affected
- Biases present in training data may lead to unfair treatment of certain groups

Answer: Accountability; opacity/difficulty explaining decisions; bias and unfair treatment

2. What makes deep learning conceptually different from traditional programming approaches? (Select ALL that apply)

- Deep learning uses more computer memory than traditional programming
- Human programmers define the goal but not the specific steps to achieve it
- The system builds increasingly complex representations from simple inputs automatically
- Instead of programming explicit rules, the system discovers patterns through layers of abstraction

Answer: Human programmers define the goal but not the specific steps; the system builds increasingly complex representations; it discovers patterns through layers of abstraction

3. What does the concept of "emergence" in neural networks suggest about intelligence?

- Intelligence must be programmed explicitly by humans
- Neural networks are too simple to exhibit real intelligence
- Complex intelligent behavior can arise from simple interactions between many basic components
- Intelligence requires consciousness and self-awareness

Answer: Complex intelligent behavior can arise from simple interactions between many basic components

4. From an interdisciplinary perspective, what does machine learning reveal about the relationship between mathematics, cognition, and intelligence?

- Cognition cannot be understood through mathematical models
- Machine learning proves that mathematics is superior to human thinking
- Mathematical optimization principles may underlie both artificial and biological learning processes
- Mathematics is completely separate from how intelligence works

Answer: Mathematical optimization principles may underlie both artificial and biological learning processes

5. What fundamental assumptions about learning and knowledge does machine learning challenge? (Select ALL that apply)

- That intelligence requires understanding in the way humans experience it

- That computers should only be used for mathematical calculations
- That learning necessarily involves conscious awareness of the learning process
- That knowledge must be explicitly taught rather than discovered from data

Answer: Intelligence requires human-like understanding; learning necessarily involves conscious awareness; knowledge must be explicitly taught rather than discovered from data

6. What societal implications arise from machines that can learn and adapt? (Select ALL that apply)

- Educational systems may need to reimagine what knowledge and skills are most valuable
- Questions arise about human agency and decision-making in an AI-assisted world
- Machines will only perform simple, repetitive tasks
- The nature of work and human expertise may fundamentally change

Answer: Education may need to reimagine valuable knowledge/skills; questions arise about human agency; work and expertise may fundamentally change

7. What does the success of deep learning in pattern recognition suggest about the nature of perception and understanding?

- Machines can never truly understand what they perceive
- Understanding requires emotional and subjective experience
- Human perception is fundamentally different from any computational process
- Pattern recognition may be a more fundamental aspect of intelligence than previously thought

Answer: Pattern recognition may be a more fundamental aspect of intelligence than previously thought

Recovered concept from another Quiz 2 item

Recovered answer: Humans can learn meaningful concepts from very few examples, while machines typically need thousands or millions.

Concept to study: Sample efficiency. The exact original question wording was not available in the retrieved history excerpt.

Recovered Quiz 3: Most Recent Quiz

Status: Ten questions were provided from the most recent quiz. The wording, answer choices, and correct selections below are preserved from that quiz record.

1. What fundamental philosophical challenge does distributed AI pose about the nature of intelligence and consciousness?

- Whether collective intelligence can emerge from separate agents without centralized consciousness
- Distributed systems are too slow to be considered intelligent
- Multiple AI agents cannot communicate effectively
- Distributed AI requires too much computational power

  Answer: Whether collective intelligence can emerge from separate agents without centralized consciousness

2. What societal implications arise from having AI systems distributed across multiple organizations and platforms? (Select ALL that apply)

- Accountability becomes harder to establish when problems occur
- Questions about data ownership and privacy become more complex
- Distributed systems always provide better security
- Power dynamics shift as no single entity controls the entire AI ecosystem

  Answer: Accountability becomes harder to establish; questions about data ownership and privacy become more complex; power dynamics shift as no single entity controls the entire AI ecosystem

3. Why do AI frameworks represent a fundamental shift in how we approach problem-solving? (Select ALL that apply)

- They enable rapid experimentation and iteration in AI development
- They standardize approaches across different domains and applications
- They require advanced programming skills from all users
- They democratize access to sophisticated AI capabilities for non-experts

  Answer: They enable rapid experimentation and iteration; standardize approaches across domains and applications; democratize access to sophisticated AI capabilities for non-experts

4. What does the success of natural language processing reveal about the relationship between human communication and computational understanding?

- Machines understand language exactly the same way humans do
- Natural language is too complex for machines to process effectively
- Only formal languages can be processed by computers
- Language patterns can be captured statistically without deep semantic understanding

  Answer: Language patterns can be captured statistically without deep semantic understanding

5. What cultural and linguistic challenges does global NLP deployment raise? (Select ALL that apply)

- Cultural context significantly affects meaning and interpretation
- Translation between languages can lose nuanced meaning and cultural significance
- Dominant languages may receive more development resources, creating inequality
- All languages have identical grammatical structures

Answer: Cultural context affects meaning; translation can lose nuance and cultural significance; dominant languages may receive more resources, creating inequality

6. From an interdisciplinary perspective, what does the emergence of large language models suggest about knowledge and understanding?

- Language models can never achieve human-level understanding
- Knowledge must be explicitly programmed into systems
- True understanding requires conscious experience
- Sophisticated language use might emerge from pattern recognition without explicit knowledge representation

Answer: Sophisticated language use might emerge from pattern recognition without explicit knowledge representation

7. What ethical concerns arise when AI systems can generate human-like text and speech? (Select ALL that apply)

- Questions about authorship and intellectual property in AI-generated content
- The potential for creating convincing misinformation and propaganda
- AI-generated text is always clearly distinguishable from human writing
- The possibility of deception about whether communication is coming from humans or machines

Answer: Authorship and intellectual property questions; convincing misinformation and propaganda; deception about whether communication comes from humans or machines

8. What does the concept of "emergent behavior" in distributed AI systems reveal about complex problem-solving?

- Distributed systems are less effective than centralized ones
- Complex problems always require centralized control
- Sophisticated solutions can arise from simple agents following basic rules
- Individual agents must be highly sophisticated

Answer: Sophisticated solutions can arise from simple agents following basic rules

9. How do AI frameworks change the relationship between domain expertise and technical implementation? (Select ALL that apply)

- Technical expertise becomes completely unnecessary
- The barrier between conceptual understanding and practical application is lowered
- Domain experts can focus more on problem definition rather than technical implementation
- Interdisciplinary collaboration becomes more feasible and productive

Answer: The barrier between conceptual understanding and practical application is lowered; domain experts can focus more on problem definition; interdisciplinary collaboration becomes more feasible and productive

10. What fundamental question about human nature does advanced NLP raise regarding what makes human communication unique?

- Human communication is always more accurate than machine communication
- Humans have larger vocabularies than AI systems

- Whether meaning and intention are necessary for effective communication, or if pattern matching is sufficient
- Humans are faster at processing language than machines

  Answer: Whether meaning and intention are necessary for effective communication, or if pattern matching is sufficient

Latest Quiz: Fast Recognition Rules

- Watch for absolutes such as always, never, all, identical, completely, and cannot. They were repeatedly used in distractors.
- Distributed intelligence: the group-level result can emerge without one central conscious controller.
- Distribution changes governance as well as architecture: accountability, privacy, ownership, security exposure, and power may cross organizational boundaries.
- AI frameworks lower implementation barriers but do not eliminate the need for technical expertise.
- NLP can produce effective language behavior from learned statistical patterns without settling whether the system deeply understands meaning.
- Global NLP must account for cultural context, translation loss, grammatical diversity, and unequal resources across languages.
- Human-like generation raises authorship/IP, misinformation, and source-deception concerns.

What the Quiz Pattern Suggests to Study

The recovered questions give a useful picture of the professor's question style. The emphasis is not just vocabulary. Several questions ask you to distinguish a conceptual statement from a plausible-sounding distractor, and Select ALL that apply items require evaluating every choice independently.

- Names/history: Turing, Dartmouth, AI Winter.
- Paradigms: symbolic AI, machine learning, neural networks.
- Knowledge representation: semantic networks and frames.
- Human vs. AI strengths and differences in learning.
- Deep learning vs. explicit programming.
- Ethics: accountability, explainability/opacity, and bias/fairness.
- Emergence, pattern recognition, and interdisciplinary connections to cognition.
- Broader implications for agency, education, work, and expertise.
- Core ML concepts from the lectures even when not present in recovered questions: supervised/unsupervised/reinforcement learning, generalization, overfitting, model evaluation.
- Distributed AI, AI frameworks, NLP, global language challenges, LLMs, authorship/IP, misinformation, and human–machine communication.

Fast Discrimination Rules

Semantic network	Think graph: concepts = nodes, relationships = links.
Frame	Think structured package: slots, attributes, defaults.
Symbolic AI	Explicit knowledge and rules.
Machine learning	Patterns learned from data.
Deep learning	Layered learned representations, not hand-coded step-by-step rules.
Overfitting	Great on training data, weaker on unseen data.

Abduction	Best explanation for what you observe.
Deduction	What must follow if the premises are true.
Induction	Generalization from examples.
Emergence	Complex behavior from many simpler interacting components.
Sample efficiency	How much experience/data is needed to learn a useful concept.

Mock Exam Blueprint

Use this blueprint with a practice-question generator. It is a study recommendation, not a prediction of the professor's exact topic distribution.

Topic	Questions in a 30-question practice set
History / foundations / paradigms	4
Knowledge representation + reasoning	4
Machine-learning foundations	5
Neural networks / deep learning	5
Distributed AI + frameworks	5
NLP / LLMs / language and culture	4
Ethics, society, human vs. AI	3

Prompt for a Mock Exam Generator

You can give Claude this instruction:

> *Create a 30-question ALY 3510 midterm practice exam. Use only the course concepts and recovered quiz questions I provide. Match the style of all three recovered quizzes: four answer choices, a mix of single-answer and "Select ALL that apply" questions, plausible conceptual distractors, interdisciplinary wording, and careful use of absolute terms such as always, never, all, and completely. Cover AI history, knowledge representation, reasoning, machine learning, neural networks, distributed AI, AI frameworks, NLP, LLMs, ethics, culture, and human–machine understanding. Do not repeat the recovered questions verbatim. Do not reveal the answers until I submit all 30 responses. Score the exam out of 100, explain every missed question, identify the concept behind each miss, and then generate a short targeted retest covering only weak areas with different wording.*

Final Readiness Checklist

☐ I can answer all recovered quiz questions without looking at the answer key.

☐ I can explain why each wrong answer is wrong, not just recognize the correct choice.

☐ I can distinguish semantic networks from frames.

☐ I know Turing, Dartmouth, and the causes of AI Winter.

☐ I can distinguish symbolic AI, machine learning, and neural networks.

☐ I can explain supervised, unsupervised, and reinforcement learning.

☐ I can explain overfitting and generalization.

☐ I can explain what makes deep learning different from traditional programming.

☐ I understand emergence and sample efficiency.

☐ I can identify accountability, opacity/explainability, and bias/fairness concerns.

☐ I can explain the interdisciplinary links among ML, cognition, pattern recognition, and intelligence.

☐ I can explain emergence in both neural networks and distributed multi-agent systems.

☐ I can distinguish the benefits of AI frameworks from the false claim that technical expertise becomes unnecessary.

☐ I can explain what NLP and LLM performance does and does not prove about understanding.

☐ I can identify cultural, linguistic, authorship, misinformation, privacy, accountability, and governance concerns.

☐ I have completed the Respondus practice quiz and system/webcam checks on the computer/location I plan to use.